

Network Security Assessment Project Charter

1. Project Title

Network Security Assessment

2. Project Background

Organizations rely on network infrastructure to support business operations and exchange critical information. Securing network environments is essential to protect systems, applications, and sensitive data from unauthorized access, vulnerabilities, and potential security threats.

This project focuses on performing a network security assessment to identify potential weaknesses within a network environment, evaluate associated risks, and recommend appropriate security improvements.

3. Project Purpose

The purpose of this project is to demonstrate an IT audit approach for assessing network security controls, identifying vulnerabilities, and evaluating the effectiveness of security measures.

The assessment aims to provide visibility into possible network weaknesses and recommend actions to improve confidentiality, integrity, and availability of information systems.

4. Project Objectives

The main objectives of this assessment are:

- Identify active hosts and network services.
 - Perform network scanning to understand the security posture.
 - Identify potential vulnerabilities and misconfigurations.
 - Assess security risks associated with discovered findings.
 - Recommend appropriate security controls and improvements.
 - Document findings using an audit-style reporting approach.
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5. Audit Scope

The assessment covers:

Included Areas

- Network host discovery
- Open port identification
- Service and version detection
- Vulnerability identification
- Risk assessment
- Security recommendations

Excluded Areas

- Exploitation of vulnerabilities
 - Unauthorized access attempts
 - Modification of systems
 - Production environment testing
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6. Assessment Methodology

The assessment follows a structured security review methodology:

Phase 1: Planning

- Define assessment objectives.
- Identify assessment scope.
- Select appropriate testing approach.

Phase 2: Network Scanning

- Perform network discovery.
- Identify available services and ports.
- Collect technical information.

Phase 3: Vulnerability Analysis

- Review identified services.
- Analyze potential security weaknesses.
- Categorize findings based on risk level.

Phase 4: Risk Evaluation

- Determine likelihood and impact.
- Assign risk ratings.
- Develop recommendations.

Phase 5: Reporting

- Document findings.
 - Provide remediation recommendations.
 - Prepare final assessment report.
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7. Tools and Technologies

The assessment uses:

Tool	Purpose
Nmap	Network discovery and port scanning
CVE Database	Vulnerability reference and analysis
Microsoft Excel	Risk assessment and documentation
Microsoft Word	Audit reporting

8. Project Deliverables

The project deliverables include:

- Network Security Assessment Report
 - Scan Methodology Document
 - Vulnerability Findings Report
 - Risk Matrix
 - Recommendations Report
 - Project README documentation
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9. Expected Outcomes

The expected outcomes of this project are:

- Improved understanding of network security assessment processes.
 - Identification of potential security weaknesses.
 - Better understanding of vulnerability management.
 - Development of practical IT audit documentation skills.
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10. Risks and Limitations

Risks

- Incomplete visibility of network infrastructure.
- False positive vulnerability findings.
- Limited information about system configurations.

Limitations

- Assessment is performed in a controlled environment.
 - No exploitation activities are conducted.
 - Findings are based on available assessment data.
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11. Conclusion

The Network Security Assessment project demonstrates the application of IT audit and cybersecurity principles to evaluate network security controls. The assessment process helps organizations identify weaknesses, understand associated risks, and implement appropriate security improvements.